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ECGR 4090 – Real time AI
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Homework 1

GitHub: https://github.com/ric3b0wl/ECGR4090-RealTimeAI

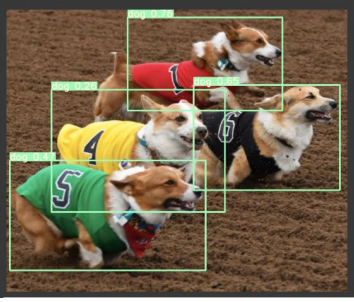
Problem 1 Run Yolo.v5 across all four different model sizes (s, m, l, x)

https://github.com/ric3b0wl/ECGR4090-RealTimeAI/tree/main/hw1

Yolo v5 small

```
start_s = time.time()

!python detect.py --weights yolov5s.pt --img 640 --conf 0.25 --source '/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/corgi.jpg' --project='/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/'
# python3 detect.py --weights yolov5s.pt --img 640 --conf 0.25 --source '/home/xam/Pictures/corgi.jpg' --project='/home/xam/Pictures/exp'

stop_s = time.time()
durtation_s = stop_s - start_s
print(f'\n Duration(small): {durtation_s} sec\n')

Image(filename='/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/exp/corgi.jpg', width=600)
```

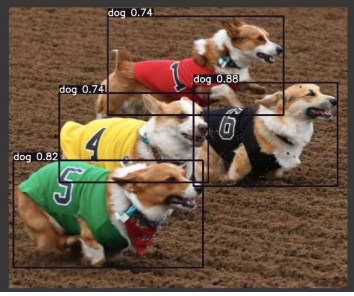
```
Namespace(agnostic_nms=False, augment=False, classes=None, conf_thres=0.25, device='', exist_ok=False, img_size=640, iou_thres=0.45, name='exp', project='/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/', save_conf=False, save_txt=False, source='/content
YOLOv5 v4.0-113-gfe6ebb9 torch 1.7.1+cu101 CPU

Fusing layers...
Model Summary: 224 layers, 7266973 parameters, 0 gradients, 17.0 GFLOPS
image 1/1 /content/drive/My Drive/ECGR4090-RealTimeAI/hw1/corgi.jpg: 544x640 4 dogs, Done. (0.462s)
Results saved to /content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/exp
Done. (0.543s)

 Duration(small): 3.632797241210937 5 sec
```



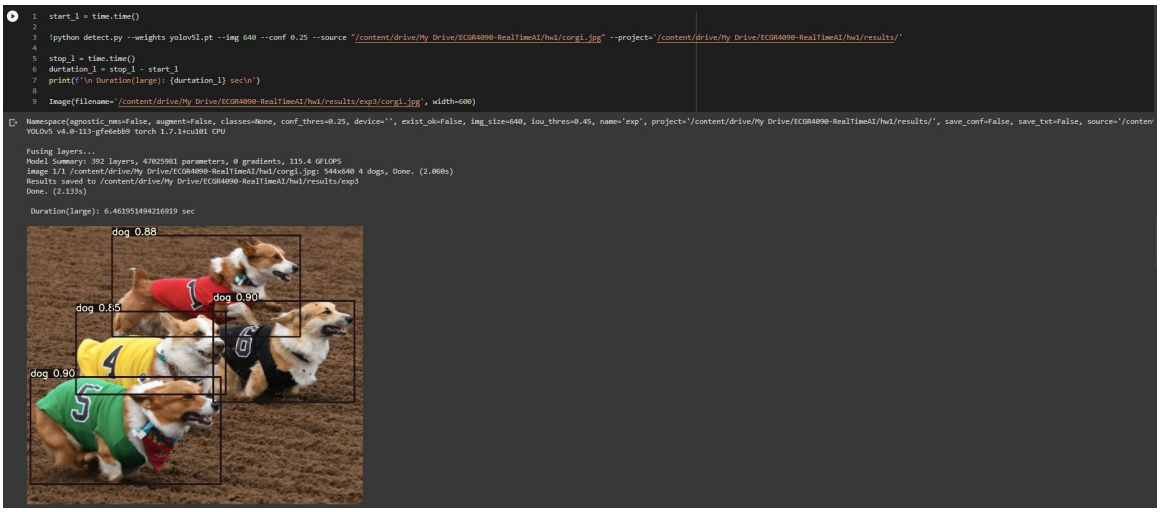
Yolo v5 medium

```
start_m = time.time()

!python detect.py --weights yolov5m.pt --img 640 --conf 0.25 --source "/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/corgi.jpg" --project='/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/'

stop_m = time.time()
durtation_m = stop_m - start_m
print(f'\n Duration(medium): {durtation_m} sec\n')

Image(filename='/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/exp2/corgi.jpg', width=600)
```

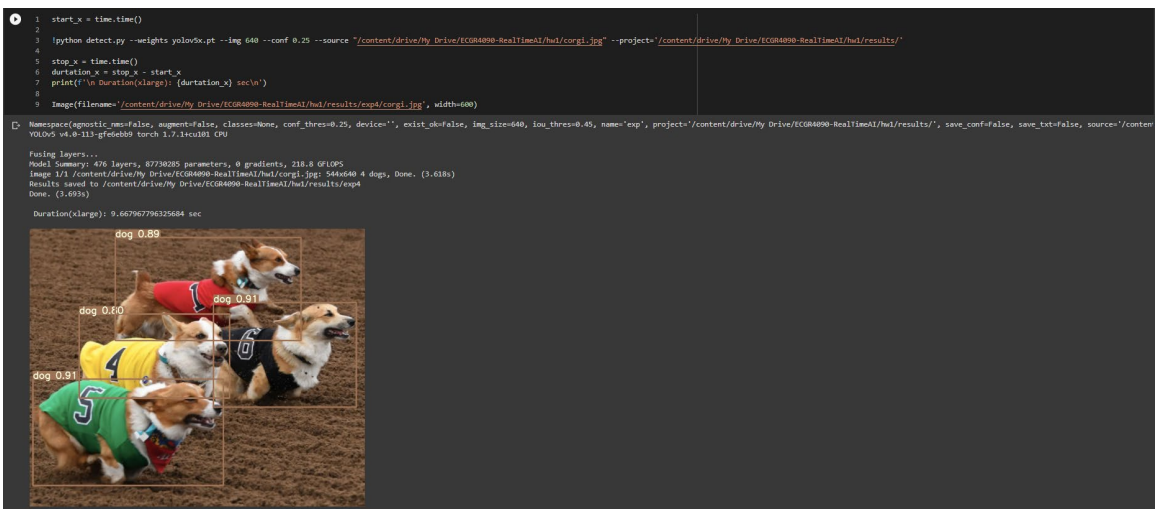
```
Namespace(agnostic_nms=False, augment=False, classes=None, conf_thres=0.25, device='', exist_ok=False, img_size=640, iou_thres=0.45, name='exp', project='/content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/', save_conf=False, save_txt=False, source='/content
YOLOv5 v4.0-113-gfe6ebb9 torch 1.7.1+cu101 CPU

Fusing layers...
Model Summary: 308 layers, 21356877 parameters, 0 gradients, 51.3 GFLOPS
image 1/1 /content/drive/My Drive/ECGR4090-RealTimeAI/hw1/corgi.jpg: 544x640 4 dogs, Done. (1.047s)
Results saved to /content/drive/My Drive/ECGR4090-RealTimeAI/hw1/results/exp2
Done. (1.120s)

 Duration(medium): 4.552726984024048 sec
```



Yolo v5 large



Yolo v5 extra large



Summary

The predication gets more and more accurate after each size model changes. Example on the red number 1 corgi, from 76% to 89% of confident rate as identify as “dog”. The downside will be because of increase amount of model size, duration time also increased from 3.63 sec to 9.66 sec.

Another example of NYC intersection street view:

Yolo v5 small

Yolo v5 medium

Yolo v5 large

Yolo v5 extra large

The same result as above corgi pictures, the confidence level increase. Because the example image above has a lot of objects in single image, it took much larger time for program to identify the objects in the images.

Problem 2 Yolo v5 on Jetson Nano 2GB

Using 4 different pretrained size yolo models on Jetson Nano

I was not able to run none of extra larger size model on jetson Nano 2GB, it always killed itself after running out of resource.

There are some examples of result in single batch:

```
Model Summary: 224 layers, 7266973 parameters, 0 gradients, 17.0 GFLOPS
image 1/1 /home/sam/Pictures/corgi.jpg: 544x640 4 dogs, Done. (3.235s)
Results saved to /home/sam/Pictures/exp/exp
Done. (10.898s)
```

Single batch in small size

```
Model Summary: 308 layers, 21356877 parameters, 0 gradients, 51.3 GFLOPS
image 1/1 /home/sam/Pictures/img0.jpg: 448x640 2 zebras, 1 giraffe, Done. (1.683s)
Results saved to /home/sam/Pictures/exp2
Done. (5.786s)
```

Single batch in medium size

```
Model Summary: 392 layers, 47025981 parameters, 0 gradients, 115.4 GFLOPS
image 1/1 /home/sam/Pictures/img0.jpg: 448x640 2 zebras, 1 giraffe, Done. (2.568s)
Results saved to /home/sam/Pictures/exp2
Done. (5.137s)
```

Single batch in large size

Tabulated Results:

(all unit in second)	Small	Medium	Large	Extra Large
Single batch	3.235	1.683	2.568	DNF
4 batches	2.630	3.438	9.137	DNF
8 batches	4.762	8.399	11.527	DNF
16 batches	5.238	9.379	45.714	DNF

DNF = Did not finish

As above can show the time different between batches size and model size. When either batch or model size increase, it takes longer time to finish object detections. I also found it is interesting that time different between each image is not determine how many objects in the image, but it depends on the confident level of object in a single image.

Problem 2 Yolo v5 on Jetson Nano 2GB Real time camera detection

Demo video link: https://youtu.be/raz6AccXkZc

GitHub code: https://github.com/ric3b0wl/ECGR4090-RealTimeAI/blob/main/hw1/detect_jetcam.py

```
import torch
from PIL import Image
from matplotlib import cm
from jetcam.usb_camera import USBCamera
from IPython.display import display
from jetcam.utils import bgr8_to_jpeg

model = torch.hub.load('ultralytics/yolov5', 'yolov5s', pretrained=True)

webcam = USBCamera(width=224, height=224, capture_width=640, capture_height=480,
capture_device=0)

image = webcam.read()
# print(webcam.shape)
# print(webcam.value.shape)

webcam.running = True

def update_image(change):
    image = change['new']
    im = Image.fromarray(image)
    im.show()
    result = model(image)
    result.print()
    im.close()

webcam.observe(update_image, names='value')
```